

From Second Quantization and Quasispin to Pauli Strings

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Scope of these notes

These notes explain how to rewrite many-body Hamiltonians in terms of Pauli strings built from

$$I, \quad X, \quad Y, \quad Z,$$

so that they are ready for quantum-computing implementations.

Main topics:

- the Lipkin Hamiltonian in quasispin form,
- detailed mappings to Pauli operators,
- compressed and direct qubit encodings,
- second-quantized fermionic operators $a^\dagger, a$,
- the nuclear pairing Hamiltonian as a second worked example,
- a fuller Jordan–Wigner mapping from fermions to qubits.

Notation

We use:

- $a_p^\dagger$: fermion creation operator in orbital p ,
- a_p : fermion annihilation operator in orbital p ,
- $n_p = a_p^\dagger a_p$: number operator,
- X_i, Y_i, Z_i : Pauli operators on qubit i ,
- I_i : identity on qubit i .

Throughout, we use only

$$a_p^\dagger, \quad a_p.$$

For quasispin systems we use collective operators

$$J_x, J_y, J_z, \quad J_\pm = J_x \pm iJ_y.$$

For pairing we use local quasispin operators

$$S_p^\pm, \quad S_p^z.$$

Why Pauli decompositions matter

Quantum algorithms typically require a Hamiltonian of the form

$$H = \sum_{\alpha} h_{\alpha} P_{\alpha}, \quad P_{\alpha} \in \{I, X, Y, Z\}^{\otimes n}.$$

Once a Hamiltonian is written in this way, one can:

- estimate energies in VQE,
- generate Trotterized time evolution,
- build QAOA circuits,
- exploit commutativity and measurement grouping.

So the main preprocessing step is:

rewrite the many-body Hamiltonian as a weighted sum of Pauli strings.

Lipkin Hamiltonian

We consider the Lipkin Hamiltonian in quasispin representation

$$H = \frac{\epsilon}{2} \left(J_z - \frac{N}{2} \right) + \frac{V}{2} (J_+^2 + J_-^2) + \frac{W}{2} J_z^2.$$

Here:

- N is the number of particles,
- ϵ sets the one-body splitting,
- V controls pair scattering between the two levels,
- W weights the quadratic J_z^2 term.

The collective operators satisfy the $su(2)$ algebra

$$[J_x, J_y] = iJ_z, \quad [J_y, J_z] = iJ_x, \quad [J_z, J_x] = iJ_y,$$

or, equivalently,

$$[J_z, J_{\pm}] = \pm J_{\pm}, \quad [J_+, J_-] = 2J_z.$$

The basis states are

$$|j, m\rangle, \quad m = -j, -j+1, \dots, j,$$

with

$$J_z |j, m\rangle = m |j, m\rangle.$$

Action of ladder operators

The ladder operators act as

$$J_+ |j, m\rangle = \sqrt{j(j+1) - m(m+1)} |j, m+1\rangle,$$

$$J_- |j, m\rangle = \sqrt{j(j+1) - m(m-1)} |j, m-1\rangle.$$

Therefore

$$J_+^2 |j, m\rangle = \sqrt{j(j+1) - m(m+1)} \sqrt{j(j+1) - (m+1)(m+2)} |j, m+2\rangle,$$

and analogously for J_-^2 .

Useful identity for the interaction term

Using

$$J_{\pm} = J_x \pm iJ_y,$$

we compute

$$J_+^2 + J_-^2 = (J_x + iJ_y)^2 + (J_x - iJ_y)^2 = 2(J_x^2 - J_y^2).$$

Hence the Lipkin Hamiltonian can also be written as

$$H = \frac{\epsilon}{2} \left(J_z - \frac{N}{2} \right) + V(J_x^2 - J_y^2) + \frac{W}{2} J_z^2.$$

Two mapping strategies

There are two main strategies for turning the Lipkin Hamiltonian into Pauli strings:

- 1 **Compressed encoding of the quasispin basis:** encode the $(2j + 1)$ -dimensional quasispin space into $\lceil \log_2(2j + 1) \rceil$ qubits.
- 2 **Direct collective-spin mapping:** write

$$J_\alpha = \frac{1}{2} \sum_i \sigma_i^\alpha$$

on a multi-qubit register and expand the Hamiltonian.

We will use:

- compressed encoding for $N = 4$,
- direct collective mapping for $N = 6$.

Four particles: quasispin dimension and encoding

For $N = 4$,

$$j = \frac{N}{2} = 2, \quad 2j + 1 = 5.$$

Thus the quasispin sector has dimension 5, which fits into 3 qubits since

$$2^2 = 4 < 5 \leq 8 = 2^3.$$

We choose the binary map

$$|2, -2\rangle \leftrightarrow |000\rangle, \quad |2, -1\rangle \leftrightarrow |001\rangle, \quad |2, 0\rangle \leftrightarrow |010\rangle,$$

$$|2, 1\rangle \leftrightarrow |011\rangle, \quad |2, 2\rangle \leftrightarrow |100\rangle.$$

The states $|101\rangle, |110\rangle, |111\rangle$ are unphysical.

Projectors and transition operators on qubits

For one qubit:

$$\begin{aligned} |0\rangle\langle 0| &= \frac{I + Z}{2}, & |1\rangle\langle 1| &= \frac{I - Z}{2}, \\ |1\rangle\langle 0| &= \frac{X - iY}{2}, & |0\rangle\langle 1| &= \frac{X + iY}{2}. \end{aligned}$$

Therefore:

- diagonal operators are expanded with I and Z ,
- off-diagonal transitions are expanded with X and Y ,
- any finite operator in the encoded space can be written as a sum of Pauli strings.

Three-qubit form of J_z

Since

$$J_z = \sum_{m=-2}^2 m |m\rangle \langle m|,$$

we obtain

$$J_z = -2 |000\rangle \langle 000| - |001\rangle \langle 001| + |011\rangle \langle 011| + 2 |100\rangle \langle 100|.$$

Expanding the projectors gives

$$J_z = -\frac{1}{4} IZI + \frac{1}{4} IZZ - \frac{1}{2} ZII - \frac{1}{2} ZIZ - \frac{3}{4} ZZI - \frac{1}{4} ZZZ.$$

Three-qubit form of J_z^2

Similarly,

$$J_z^2 = \sum_{m=-2}^2 m^2 |m\rangle \langle m|,$$

which yields

$$J_z^2 = \frac{5}{4} III + \frac{3}{4} IIZ + IZI + IZZ + \frac{1}{4} ZII - \frac{1}{4} ZIZ.$$

These formulas are exact on the physical 5-dimensional subspace.

Three-qubit form of the ladder terms

For $j = 2$,

$$\begin{aligned} J_+ |-2\rangle &= 2 |-1\rangle, & J_+ |-1\rangle &= \sqrt{6} |0\rangle, \\ J_+ |0\rangle &= \sqrt{6} |1\rangle, & J_+ |1\rangle &= 2 |2\rangle. \end{aligned}$$

Therefore

$$\begin{aligned} J_+^2 + J_-^2 &= \frac{3 + \sqrt{6}}{2} IXI + \frac{\sqrt{6} - 3}{2} IXZ + \frac{\sqrt{6}}{2} XXI + \frac{\sqrt{6}}{2} XXZ \\ &+ \frac{\sqrt{6}}{2} YYI + \frac{\sqrt{6}}{2} YYZ + \frac{3 + \sqrt{6}}{2} ZXI + \frac{\sqrt{6} - 3}{2} ZXZ. \end{aligned}$$

Final $N = 4$, 3-qubit Lipkin Hamiltonian

Substituting into

$$H = \frac{\epsilon}{2}(J_z - 2) + \frac{V}{2}(J_+^2 + J_-^2) + \frac{W}{2}J_z^2$$

gives

$$\begin{aligned} H_{N=4}^{(3q)} = & \left(-\epsilon + \frac{5W}{8}\right) III + \frac{3W}{8} IIZ + \left(-\frac{\epsilon}{8} + \frac{W}{2}\right) IZI + \left(\frac{\epsilon}{8} + \frac{W}{2}\right) IZZ \\ & + \left(-\frac{\epsilon}{4} + \frac{W}{8}\right) ZII + \left(-\frac{\epsilon}{4} - \frac{W}{8}\right) ZIZ - \frac{3\epsilon}{8} ZZI - \frac{\epsilon}{8} ZZZ \\ & + \frac{V(3 + \sqrt{6})}{4} IXI + \frac{V(\sqrt{6} - 3)}{4} IXZ + \frac{V\sqrt{6}}{4} XXI + \frac{V\sqrt{6}}{4} XXZ + \frac{V\sqrt{6}}{4} YYI + \frac{V\sqrt{6}}{4} YYZ + \frac{V(3 + \sqrt{6})}{4} ZII \\ & + \frac{V(\sqrt{6} - 3)}{4} ZIZ + \frac{V\sqrt{6}}{4} ZZI + \frac{V\sqrt{6}}{4} ZZZ \end{aligned}$$

Because the 3-qubit register has 8 basis states but only 5 are physical, one often adds

$$H_{\text{pen}} = \lambda P_{\text{unphys}},$$

where

$$P_{\text{unphys}} = |101\rangle \langle 101| + |110\rangle \langle 110| + |111\rangle \langle 111|.$$

This energetically suppresses leakage outside the encoded quasispin space.

Six particles: direct collective-spin realization

For $N = 6$, a simple and transparent mapping uses six qubits and defines

$$J_x = \frac{1}{2} \sum_{i=1}^6 X_i, \quad J_y = \frac{1}{2} \sum_{i=1}^6 Y_i, \quad J_z = \frac{1}{2} \sum_{i=1}^6 Z_i.$$

This is not qubit-optimal, since the symmetric quasispin sector has dimension 7, but it leads to a very clean Pauli-string Hamiltonian.

Expanding the collective squares

Using $X_i^2 = Y_i^2 = Z_i^2 = I$,

$$J_x^2 = \frac{1}{4} \sum_{i=1}^6 I + \frac{1}{4} \sum_{i \neq j} X_i X_j = \frac{6}{4} I + \frac{1}{2} \sum_{i < j} X_i X_j,$$

and similarly

$$J_y^2 = \frac{6}{4} I + \frac{1}{2} \sum_{i < j} Y_i Y_j,$$

$$J_z^2 = \frac{6}{4} I + \frac{1}{2} \sum_{i < j} Z_i Z_j.$$

Interaction term on 6 qubits

Using

$$J_+^2 + J_-^2 = 2(J_x^2 - J_y^2),$$

we get

$$\frac{V}{2}(J_+^2 + J_-^2) = V(J_x^2 - J_y^2) = \frac{V}{2} \sum_{i < j} (X_i X_j - Y_i Y_j).$$

Thus the interaction is an all-to-all two-body Pauli interaction.

Linear and quadratic J_z terms on 6 qubits

For the linear term,

$$\frac{\epsilon}{2} \left(J_z - \frac{N}{2} \right) = \frac{\epsilon}{2} \left(\frac{1}{2} \sum_{i=1}^6 Z_i - 3I \right) = \frac{\epsilon}{4} \sum_{i=1}^6 Z_i - \frac{3\epsilon}{2} I.$$

For the quadratic term,

$$\frac{W}{2} J_z^2 = \frac{W}{2} \left(\frac{6}{4} I + \frac{1}{2} \sum_{i < j} Z_i Z_j \right) = \frac{3W}{4} I + \frac{W}{4} \sum_{i < j} Z_i Z_j.$$

Final $N = 6$, 6-qubit Lipkin Hamiltonian

Collecting all terms gives

$$H_{N=6}^{(6q)} = \left(-\frac{3\epsilon}{2} + \frac{3W}{4} \right) I + \frac{\epsilon}{4} \sum_{i=1}^6 Z_i + \frac{V}{2} \sum_{i<j} (X_i X_j - Y_i Y_j) + \frac{W}{4} \sum_{i<j} Z_i Z_j.$$

This is already directly usable in qubit-based quantum algorithms.

Comparison of the two Lipkin mappings

Case	Qubits	Main feature
$N = 4$, compressed	3	minimal encoding
$N = 6$, direct	6	simplest Pauli structure

Compressed mappings save qubits but require code-space control. Direct mappings use more qubits but give very transparent Hamiltonians.

Second quantization for fermions

In fermionic many-body theory, the basic operators are

$$a_p^\dagger, \quad a_p,$$

satisfying anticommutation relations

$$\{a_p, a_q^\dagger\} = \delta_{pq}, \quad \{a_p, a_q\} = 0, \quad \{a_p^\dagger, a_q^\dagger\} = 0.$$

A general second-quantized Hamiltonian has the form

$$H = \sum_{pq} h_{pq} a_p^\dagger a_q + \frac{1}{4} \sum_{pqrs} v_{pqrs} a_p^\dagger a_q^\dagger a_s a_r.$$

Occupation-number basis and qubits

A fermionic Fock basis state is specified by occupations

$$|n_0 n_1 \cdots n_{M-1}\rangle, \quad n_p \in \{0, 1\}.$$

This is already a binary basis, so one may map it directly to M qubits:

$$|n_0 n_1 \cdots n_{M-1}\rangle_{\text{Fock}} \leftrightarrow |n_0 n_1 \cdots n_{M-1}\rangle_{\text{qubit}}.$$

The nontrivial part is not the basis mapping but the operator mapping: fermionic signs must be reproduced correctly on qubits.

Why fermion-to-qubit mappings are needed

A quantum computer acts on qubits, not directly on fermionic Fock states. So we need a mapping

$$a_p^\dagger, a_p, n_p = a_p^\dagger a_p \longrightarrow \text{Pauli strings.}$$

The most common constructions are:

- Jordan–Wigner,
- Bravyi–Kitaev,
- parity mapping.

We focus here on Jordan–Wigner, since it is the most direct mathematically.

Deriving the Jordan–Wigner idea

For a single qubit,

$$\sigma^+ = \frac{X - iY}{2} = |1\rangle\langle 0|, \quad \sigma^- = \frac{X + iY}{2} = |0\rangle\langle 1|.$$

If fermions were distinguishable, we could simply identify

$$a_p^\dagger \sim \sigma_p^+, \quad a_p \sim \sigma_p^-.$$

But this fails because different fermionic modes must anticommute:

$$a_p a_q = -a_q a_p \quad (p \neq q).$$

So we must insert a parity string that counts the number of occupied orbitals before site p .

Jordan–Wigner transformation

For orbital p , the Jordan–Wigner mapping is

$$a_p^\dagger = \left(\prod_{j=0}^{p-1} Z_j \right) \frac{X_p - iY_p}{2}, \quad a_p = \left(\prod_{j=0}^{p-1} Z_j \right) \frac{X_p + iY_p}{2}.$$

The string of Z 's ensures the correct fermionic sign structure. It is often called the *parity string*.

Why the parity string works

Recall that

$$Z|0\rangle = +|0\rangle, \quad Z|1\rangle = -|1\rangle.$$

Therefore

$$\prod_{j=0}^{p-1} Z_j$$

returns a factor

$$(-1)^{n_0+n_1+\dots+n_{p-1}}$$

on a basis state.

This is exactly the fermionic sign picked up when a creation or annihilation operator is moved through all earlier occupied orbitals in the canonical ordering.

Checking the canonical anticommutator

Using the Jordan–Wigner map, one can verify

$$a_p a_p^\dagger + a_p^\dagger a_p = I.$$

Indeed, since the parity string squares to the identity,

$$a_p^\dagger a_p = \frac{1 - Z_p}{2}, \quad a_p a_p^\dagger = \frac{1 + Z_p}{2},$$

and therefore

$$a_p a_p^\dagger + a_p^\dagger a_p = \frac{1 + Z_p}{2} + \frac{1 - Z_p}{2} = I.$$

Number operators under Jordan–Wigner

A key identity is

$$n_p = a_p^\dagger a_p = \frac{1 - Z_p}{2}.$$

Thus one-body diagonal terms become sums of I and Z :

$$\sum_p \epsilon_p a_p^\dagger a_p \longrightarrow \sum_p \epsilon_p \frac{1 - Z_p}{2}.$$

This is one of the most useful simplifications in practical qubit Hamiltonians.

Nearest-neighbor hopping example

For $p < q$, one finds

$$a_p^\dagger a_q + a_q^\dagger a_p$$

maps to

$$\frac{1}{2} (X_p Z_{p+1} \cdots Z_{q-1} X_q + Y_p Z_{p+1} \cdots Z_{q-1} Y_q).$$

So hopping terms become nonlocal Pauli strings because of the fermionic parity chain.

Pair creation operator in full orbital space

Consider a time-reversed pair $(p, \bar{p})$ with $p < \bar{p}$. Then

$$P_p^\dagger = a_p^\dagger a_{\bar{p}}^\dagger.$$

Using Jordan–Wigner,

$$a_p^\dagger a_{\bar{p}}^\dagger = \left(\prod_{j=0}^{p-1} Z_j \right) \frac{X_p - iY_p}{2} \left(\prod_{j=0}^{\bar{p}-1} Z_j \right) \frac{X_{\bar{p}} - iY_{\bar{p}}}{2}.$$

After cancelling the common prefix, one gets a Pauli string with a parity chain between p and $\bar{p}$.

Explicit formula for $a_p^\dagger a_q^\dagger$

For $p < q$,

$$a_p^\dagger a_q^\dagger = \frac{1}{4} (X_p - iY_p) Z_{p+1} \cdots Z_{q-1} (X_q - iY_q).$$

Expanding this product gives

$$a_p^\dagger a_q^\dagger = \frac{1}{4} \left(X_p Z \cdots Z X_q - i X_p Z \cdots Z Y_q - i Y_p Z \cdots Z X_q - Y_p Z \cdots Z Y_q \right),$$

where the intermediate Z -string runs from $p+1$ to $q-1$.

Hermitian pair transfer operator

A very common Hermitian combination is

$$a_p^\dagger a_q^\dagger + a_q a_p.$$

Under Jordan–Wigner this becomes

$$\frac{1}{2} (X_p Z_{p+1} \cdots Z_{q-1} X_q - Y_p Z_{p+1} \cdots Z_{q-1} Y_q).$$

This is the pairing analogue of the familiar hopping identity, with a sign change between the XX and YY channels.

Local pairing operators

For each time-reversed pair $(p, \bar{p})$, define

$$P_p^\dagger = a_p^\dagger a_{\bar{p}}^\dagger, \quad P_p = a_{\bar{p}} a_p,$$

and the pair number operator

$$N_p = a_p^\dagger a_p + a_{\bar{p}}^\dagger a_{\bar{p}}.$$

Now define local quasispin operators

$$S_p^+ = P_p^\dagger, \quad S_p^- = P_p, \quad S_p^z = \frac{1}{2}(N_p - 1).$$

Commutator $[S_p^z, S_p^+]$

We derive the quasispin algebra directly from fermion operators.

Using

$$S_p^z = \frac{1}{2}(N_p - 1), \quad S_p^+ = a_p^\dagger a_{\bar{p}}^\dagger,$$

we compute

$$[S_p^z, S_p^+] = \frac{1}{2}[N_p, a_p^\dagger a_{\bar{p}}^\dagger].$$

Since

$$[N_p, a_p^\dagger] = a_p^\dagger, \quad [N_p, a_{\bar{p}}^\dagger] = a_{\bar{p}}^\dagger,$$

it follows that

$$[N_p, a_p^\dagger a_{\bar{p}}^\dagger] = 2a_p^\dagger a_{\bar{p}}^\dagger.$$

Hence

$$[S_p^z, S_p^+] = S_p^+.$$

Commutator $[S_p^z, S_p^-]$

Similarly,

$$[S_p^z, S_p^-] = \frac{1}{2}[N_p, a_{\bar{p}}a_p].$$

Using

$$[N_p, a_p] = -a_p, \quad [N_p, a_{\bar{p}}] = -a_{\bar{p}},$$

we find

$$[N_p, a_{\bar{p}}a_p] = -2a_{\bar{p}}a_p.$$

Therefore

$$[S_p^z, S_p^-] = -S_p^-.$$

Commutator $[S_p^+, S_p^-]$

Now compute

$$[S_p^+, S_p^-] = a_p^\dagger a_{\bar{p}}^\dagger a_{\bar{p}} a_p - a_{\bar{p}} a_p a_p^\dagger a_{\bar{p}}^\dagger.$$

The first term is

$$a_p^\dagger a_{\bar{p}}^\dagger a_{\bar{p}} a_p = a_p^\dagger n_{\bar{p}} a_p = n_p n_{\bar{p}}.$$

The second term becomes

$$a_{\bar{p}}(1 - n_p)a_{\bar{p}}^\dagger = (1 - n_p)(1 - n_{\bar{p}}).$$

Subtracting yields

$$[S_p^+, S_p^-] = n_p + n_{\bar{p}} - 1 = N_p - 1 = 2S_p^z.$$

Local quasispin algebra

We have proved

$$[S_p^z, S_p^\pm] = \pm S_p^\pm, \quad [S_p^+, S_p^-] = 2S_p^z.$$

So for each pair level p , the operators

$$S_p^+, \quad S_p^-, \quad S_p^z$$

generate an $su(2)$ algebra.

For different levels $p \neq q$, the operators commute:

$$[S_p^\alpha, S_q^\beta] = 0.$$

Thus a pairing model is naturally a sum of commuting local quasispins.

Collective quasispin

Defining

$$S^{\pm} = \sum_p S_p^{\pm}, \quad S^z = \sum_p S_p^z,$$

we immediately get the collective algebra

$$[S^z, S^{\pm}] = \pm S^{\pm}, \quad [S^+, S^-] = 2S^z.$$

This is the precise sense in which the pairing problem can be rewritten as a spin problem. It also explains the close analogy with the Lipkin Hamiltonian.

Nuclear pairing Hamiltonian

A standard reduced pairing Hamiltonian is

$$H_{\text{pair}} = \sum_p \epsilon_p N_p - G \sum_{p,q} P_p^\dagger P_q,$$

where

$$N_p = a_p^\dagger a_p + a_{\bar{p}}^\dagger a_{\bar{p}},$$
$$P_p^\dagger = a_p^\dagger a_{\bar{p}}^\dagger, \quad P_p = a_{\bar{p}} a_p.$$

Here p and $\bar{p}$ denote time-reversed partners.

Pairing Hamiltonian in quasispin form

Using

$$S_p^+ = P_p^\dagger, \quad S_p^- = P_p, \quad S_p^z = \frac{1}{2}(N_p - 1),$$

the pairing Hamiltonian becomes

$$H_{\text{pair}} = \sum_p \epsilon_p (2S_p^z + 1) - G \sum_{p,q} S_p^+ S_q^-.$$

Thus the pairing Hamiltonian is an interacting quasispin Hamiltonian. This is the natural starting point for compact qubit encodings.

One pair level as one qubit

For a single pair $(p, \bar{p})$, the paired subspace is two-dimensional:

$$|0\rangle_p \equiv \text{pair unoccupied}, \quad |1\rangle_p \equiv \text{pair occupied}.$$

In this reduced subspace, we may identify

$$S_p^+ = \frac{X_p - iY_p}{2}, \quad S_p^- = \frac{X_p + iY_p}{2}, \quad S_p^z = -\frac{Z_p}{2}.$$

Also,

$$N_p = 1 - Z_p.$$

Pairing Hamiltonian in pair-qubit form

Substituting

$$N_p = 1 - Z_p, \quad P_p^\dagger = \frac{X_p - iY_p}{2}, \quad P_p = \frac{X_p + iY_p}{2},$$

we obtain

$$H_{\text{pair}} = \sum_p \epsilon_p (1 - Z_p) - G \sum_{p,q} S_p^+ S_q^-.$$

Since

$$S_p^+ S_q^- + S_q^+ S_p^- = \frac{1}{2}(X_p X_q + Y_p Y_q),$$

the interaction becomes an XY -type coupling between pair qubits.

Explicit pairing Hamiltonian for Ω pair levels

For Ω pair orbitals, the pairing Hamiltonian in the paired subspace can be written as

$$H_{\text{pair}} = \sum_{p=1}^{\Omega} \epsilon_p (1 - Z_p) - \frac{G}{2} \sum_{p \neq q} (X_p X_q + Y_p Y_q) - G \sum_p S_p^+ S_p^-.$$

Since

$$S_p^+ S_p^- = \frac{1 - Z_p}{2},$$

this yields

$$H_{\text{pair}} = \sum_{p=1}^{\Omega} \left(\epsilon_p - \frac{G}{2} \right) (1 - Z_p) - \frac{G}{2} \sum_{p \neq q} (X_p X_q + Y_p Y_q).$$

Example: two pair levels in pair-qubit space

For two pair orbitals $p = 1, 2$, the paired-subspace Hamiltonian becomes

$$H_{\text{pair}}^{(2)} = \left(\epsilon_1 - \frac{G}{2} \right) (1 - Z_1) + \left(\epsilon_2 - \frac{G}{2} \right) (1 - Z_2) - \frac{G}{2} (X_1 X_2 + Y_1 Y_2).$$

This is a compact two-qubit Hamiltonian ready for variational or time-evolution algorithms.

Full orbital-space example: one pair level

Now keep the two fermionic orbitals p and $\bar{p}$ explicitly as two qubits. The one-level pairing Hamiltonian is

$$H = \epsilon_p(n_p + n_{\bar{p}}) - G a_p^\dagger a_{\bar{p}}^\dagger a_{\bar{p}} a_p.$$

Using Jordan–Wigner,

$$n_p = \frac{1 - Z_p}{2}, \quad n_{\bar{p}} = \frac{1 - Z_{\bar{p}}}{2}.$$

Also

$$a_p^\dagger a_{\bar{p}}^\dagger a_{\bar{p}} a_p = n_p n_{\bar{p}}.$$

Therefore

$$H = \frac{\epsilon_p}{2}(2 - Z_p - Z_{\bar{p}}) - \frac{G}{4}(1 - Z_p)(1 - Z_{\bar{p}}).$$

Expanded one-pair full-orbital Pauli form

Expanding the previous result gives

$$H = \left(\epsilon_p - \frac{G}{4} \right) I + \left(-\frac{\epsilon_p}{2} + \frac{G}{4} \right) Z_p + \left(-\frac{\epsilon_p}{2} + \frac{G}{4} \right) Z_{\bar{p}} - \frac{G}{4} Z_p Z_{\bar{p}}.$$

So even the full-orbital one-pair Hamiltonian reduces to a simple two-qubit diagonal Pauli Hamiltonian.

Full orbital-space example: pair transfer between two levels

Consider two pair levels p and q , with orbitals

$$(p, \bar{p}, q, \bar{q}).$$

A pairing transfer term is

$$P_p^\dagger P_q + P_q^\dagger P_p = a_p^\dagger a_{\bar{p}}^\dagger a_{\bar{q}} a_q + a_q^\dagger a_{\bar{q}}^\dagger a_{\bar{p}} a_p.$$

Under Jordan–Wigner, this becomes a Hermitian sum of four-qubit Pauli strings.

Explicit Pauli decomposition of pair transfer

Assume the orbital ordering

$$p < \bar{p} < q < \bar{q}.$$

Then one finds

$$\begin{aligned} P_p^\dagger P_q + P_q^\dagger P_p = & \frac{1}{8} \left(X_p X_{\bar{p}} X_q X_{\bar{q}} - X_p Y_{\bar{p}} Y_q X_{\bar{q}} \right. \\ & \left. - Y_p X_{\bar{p}} X_q Y_{\bar{q}} + Y_p Y_{\bar{p}} Y_q Y_{\bar{q}} \right) \end{aligned}$$

up to the parity-string conventions induced by the chosen ordering. In adjacent pair ordering, the intermediate Z -chains cancel within the paired sector, leaving a compact four-body expression.

Interpretation of the full-orbital pairing terms

This explicit decomposition shows:

- full fermionic orbital space leads to longer Pauli strings,
- pair-qubit reduction produces shorter and more structured qubit Hamiltonians,
- both are exact, but they operate in different Hilbert spaces.

So there is a practical tradeoff:

- full-space mappings are fully general,
- reduced quasispin/pair encodings are more compact when the physics is dominated by seniority-zero pairing.

Lipkin versus pairing: structural comparison

Both Hamiltonians admit spin-like descriptions:

- Lipkin:

$$H_{\text{Lipkin}} = \frac{\epsilon}{2} \left(J_z - \frac{N}{2} \right) + \frac{V}{2} (J_+^2 + J_-^2) + \frac{W}{2} J_z^2,$$

- Pairing:

$$H_{\text{pair}} = \sum_p \epsilon_p N_p - G \sum_{p,q} P_p^\dagger P_q.$$

Both become sums of Pauli strings after:

- choosing an encoding,
- expressing collective or pair operators in terms of X, Y, Z .

General recipe: from second quantization to qubits

For a fermionic Hamiltonian:

- 1 write the Hamiltonian in terms of $a_p^\dagger, a_p$,
- 2 decide whether a reduced quasispin or pair-qubit subspace is appropriate,
- 3 otherwise apply Jordan–Wigner or a related mapping,
- 4 simplify number and pair operators,
- 5 collect the resulting Pauli strings.

For collective-spin models:

- 1 write the model in J_x, J_y, J_z or ladder-operator form,
- 2 choose compressed or direct qubit encoding,
- 3 expand diagonal and off-diagonal matrix elements into Pauli strings.

Summary

In this expanded version we added:

- a fuller Jordan–Wigner derivation,
- a formal derivation of the quasispin algebra directly from second quantization,
- explicit Pauli decompositions for small pairing examples in full orbital space,
- and a more systematic comparison of compact versus full-space encodings.

The unifying message is:

fermions or collective spins $\longrightarrow$ spin algebra $\longrightarrow$ Pauli strings.

End

Questions?